

Customer Shopping Behavior Analysis Report

1. Project Overview

This project analyzes customer shopping behaviour using transactional data from 5,000 purchases across various product categories. The goal of this project was to analyze customer purchasing behavior, identify revenue-driving segments, and provide actionable business insights helps stakeholders understand customer demographics, purchasing patterns, subscription behavior, product performance, and sales trends to support strategic business decisions.

2. Dataset Summary

-**Rows:** 5050

-**Columns:** 17

-**Key Features:**

-**Customer demographics** (Age, Gender, Location, Subscription Status)

-**Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)

-**Shopping behaviour** (Discount Applied, Previous Purchases, Frequency of Purchase, Review Rating, Shipping Type)

-**Missing Data:**

556 values in Purchase Amount column

370 values in Size column

601 values in Review Rating column

548 values in Previous Purchase column

-**Duplicate Data:** Total 50 duplicate data

3. Exploratory Data Analysis using Python

Begin with data preparation and cleaning using python:

- **Data loading:** Imported the dataset using **pandas**.
- **Initial Exploration:** used **df.info()** to check structure and **df.describe ()** for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	5050.000000	5050.000000	5050	5050	5050	4494.000000	5050	4680	5050	5050	4449.000000	5050	5050	5050
unique	NaN	NaN	3	30	5	NaN	54	4	25	5	NaN	2	6	2
top	NaN	NaN	Male	Shirt	Clothing	NaN	New York	M	Black	Spring	NaN	No	Standard	No
freq	NaN	NaN	3018	320	2130	NaN	337	1981	404	1280	NaN	3462	1213	2789
mean	2519.570891	44.150495	NaN	NaN	NaN	144.765236	NaN	NaN	NaN	NaN	3.668195	NaN	NaN	NaN
std	1470.402964	15.282328	NaN	NaN	NaN	275.590101	NaN	NaN	NaN	NaN	0.865357	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	10.120000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN	NaN
25%	1252.250000	31.000000	NaN	NaN	NaN	41.000000	NaN	NaN	NaN	NaN	3.000000	NaN	NaN	NaN
50%	2499.500000	44.000000	NaN	NaN	NaN	65.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	NaN
75%	3740.750000	57.000000	NaN	NaN	NaN	89.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	5099.000000	70.000000	NaN	NaN	NaN	1499.760000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
4502.000000	5050	5050
NaN	6	7
NaN	Credit Card	Monthly
NaN	954	946
25.221901	NaN	NaN
14.521635	NaN	NaN
0.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling:** checked for null values and imputed missing values in the Size, Purchase Amount, Previous Purchase, Review Rating columns using mean, mode, median for each product category.
- **Duplicate Data Handling:** Checked for duplicate values and drop the subset of data duplicate.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Create **purchase_frequency_days** column from purchase data.
- **Database Integration:** Connected python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

Performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender:** Compared total revenue generated by Male vs. female customers.

2	Male	390964.52623835864
3	Female	352060.2931404854

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount double precision
1	4568	276.3437899543379
2	4256	734.4773493975904
3	4289	1282.44
4	4576	831.3445652173913
5	4319	685.2314705882353
6	4708	735.773625
7	4584	685.2314705882353
8	4607	557.24
9	4851	685.2314705882353
10	4845	692.11
11	4558	208.02
12	4400	885.588887887887
Total rows: 529		Query complete 00:00:0

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	avg double precision
1	Standard	340.5665663856458
2	Express	358.11591144867623

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend double precision	total_revenue double precision
1	Yes	1579	270.5797353396311	427245.4021012775
2	No	3421	165.72260214991095	566937.0219548454

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Laptop	51.75
2	Phone	51.46
3	Headphones	51.20
4	Hat	50.00
5	Watch	49.68

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	4098
2	New	97
3	Returning	805

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessor...	Jewelry	171
2	2	Accessor...	Bag	164
3	3	Accessor...	Belt	161
4	1	Clothing	Shirt	315
5	2	Clothing	Pants	171
6	3	Clothing	Blouse	171
7	1	Electroni...	Phone	171
8	2	Electroni...	Headphones	166
9	3	Electroni...	Watch	157
10	1	Footwear	Shoes	303
11	2	Footwear	Sandals	160
12	3	Footwear	Sneakers	145
13	1	Outerwear	Jacket	163
14	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

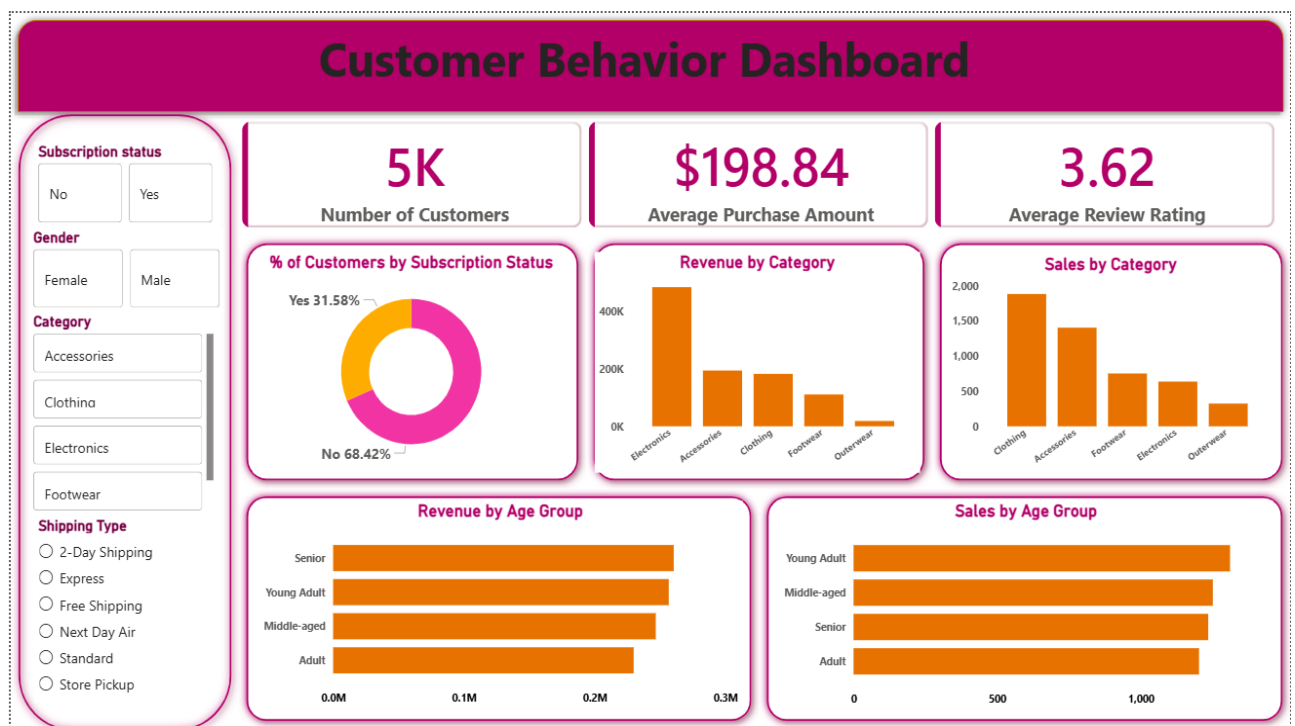
	subscription_status text	repeat_buyers bigint
1	No	2763
2	Yes	1190

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue double precision
1	Senior	260597.98704282762
2	Young Adult	256788.93997548168
3	Middle-aged	246845.8181858372
4	Adult	229949.67885197754

5. Dashboard on Power BI

Finally, built an interactive dashboard in Power BI to present insights visually.



6. Dashboard KPIs

- Total Customers: 5K

The company currently serves **5,000 customers**, providing a strong customer base for segmentation and targeted marketing.

- Average Purchase Amount: \$198.84

Customers spend nearly **\$199 per transaction**, indicating healthy purchasing power.

- Average Review Rating: 3.62 / 5

Customer satisfaction is moderate. This suggests opportunities for improving customer experience and product quality.

7. Key Insights from Dashboard

Insight 1: Subscription Adoption is Low

Observation:

- 68.42% customers are non-subscribers
- 31.58% are subscribers

Interpretation:

Most customers are not enrolled in the subscription program, indicating untapped potential for customer retention and recurring revenue.

Business Opportunity:

Subscription conversion can significantly increase customer lifetime value.

Insight 2: Electronics Generates Highest Revenue

Observation:

Electronics contributes the highest revenue (~\$500K).

Interpretation:

Although not the top-selling category by volume, electronics products have higher average transaction values.

Business Impact:

Electronics is the company's premium profit-driving category.

Insight 3: Clothing Has Highest Sales Volume

Observation:

Clothing has the highest number of sales (~1,900 units).

Interpretation:

Clothing products are high-demand but likely lower-priced than electronics.

Business Impact:

Clothing drives customer traffic and repeat engagement.

Insight 4: Outerwear Underperforms

Observation:

Outerwear shows the lowest revenue and lowest sales.

Possible Reasons:

- Weak product-market fit
- Pricing issues
- Poor marketing visibility
- Seasonal demand dependency

Insight 5: Senior Customers Generate Highest Revenue

Observation:

Senior customers contribute the highest revenue.

Interpretation:

Older customers may have greater purchasing power and willingness to spend more per transaction.

Business Opportunity:

Targeted campaigns for seniors could maximize profitability.

Insight 6: Young Adults Drive Highest Sales Count

Observation:

Young adults have the highest number of purchases.

Interpretation:

Young customers buy more frequently but may spend less per purchase.

Business Opportunity:

Potential for upselling and premium product recommendations.

Insight 7: Customer Satisfaction Needs Improvement

Observation:

Average review rating = **3.62**

Interpretation:

Customer satisfaction is moderate—not poor, but below excellence.

Risks:

- Reduced repeat purchases
- Negative reviews
- Lower brand loyalty

5. Business Recommendations / Solutions

Solution 1: Increase Subscription Enrollment

Strategies:

- Offer first-month discounts
- Loyalty reward points
- Exclusive subscriber-only deals
- Personalized email campaigns
- Free shipping for subscribers

Expected Impact:

- Higher retention
- Predictable recurring revenue
- Increased customer lifetime value

Solution 2: Expand Electronics Category

Strategies:

- Increase product variety
- Bundle accessories
- Promote premium products
- Upsell warranties/services

Expected Impact:

- Increased revenue
- Higher profit margins

Solution 3: Improve Clothing Profitability

Strategies:

- Cross-sell accessories
- Premium clothing collections
- Personalized recommendations
- Bundle discounts

Expected Impact:

Higher average order value from the highest-demand category.

Solution 4: Revive or Reassess Outerwear Category**Strategies:**

- Conduct pricing analysis
- Seasonal promotions
- Improve visibility
- Remove low-performing SKUs

Expected Impact:

Reduce inventory waste and improve category performance.

Solution 5: Target Senior Customers with Premium Campaigns**Strategies:**

- Personalized product recommendations
- VIP customer programs
- Easy shopping experience
- Customer support priority

Expected Impact:

Maximize spending from the highest-value customer segment.

Solution 6: Upsell Young Adult Customers**Strategies:**

- Bundle offers
- Premium product suggestions
- Subscription incentives
- App-based promotions

Expected Impact:

Increase average purchase value.

Solution 7: Improve Customer Satisfaction**Strategies:**

- Analyze low-rated products
- Improve shipping experience
- Enhance customer service
- Faster issue resolution
- Collect customer feedback

Expected Impact:

Higher ratings, repeat purchases, and stronger brand loyalty.